

Anomalous Behavior Detection of Marine Vessels Based on Hidden Markov Model

1st Kamran Fartash Toloue

*Department of Computer Engineering,
Mashhad Branch, Islamic Azad University,
Mashhad, Iran
kamranfartash.t@gmail.com*

2nd Majid Vafaei Jahan

*Department of Computer Engineering,
Mashhad Branch, Islamic Azad University,
Mashhad, Iran
vafaeijahan@mshdiau.ac.ir*

Abstract—In recent years, marine traffic has increased dramatically. Therefore, some vessels may show abnormal behaviors which may indicate different dangers or threats, and also decreases the marine security. These abnormal behaviors are called anomalies. Automatic detection of anomalies is one of the current issues in the maritime domain. Similar works have been proposed that often used parameters speed and location to detect anomalies, and also high false alarm rate is one of the main problems of those. In this paper, a novel approach to maritime vessels anomalous behavior detection based on Hidden Markov Model with the use of parameters speed, location and course have been proposed to increase the situational awareness. Implementation results of the proposed method represent low false alarm rate and more than 96% accuracy in the classification of normal behavior from anomalous behaviors.

Index Terms—Anomaly Detection, Situational Awareness, Maritime Surveillance, Hidden Markov Model

I. INTRODUCTION

Maritime transport accounts for approximately 90% of global trade by volume, which represents more than 50,000 vessels travelling in the oceans every day [12]. Such increase in traffic has led to a massive data overload from various information systems and sensors. Moreover, some vessels may show abnormal behaviors, which in maritime surveillance domain are called anomalies. Using different definitions for anomalies has yielded into a confusion in detection of normal from unusual behaviors [2], [9]. Hence, to give a comprehensive definition for anomalies, it can be defined as a relative event which cannot be attributed to a specific classification or category [9].

Therefore, it is necessary to design a systematic approach to automatically analyze and detect abnormal behaviors. In recent years, there has been an increasing interest in studying new approaches for marine surveillance [7]. In each approach, the quality of behavioral analysis and anomaly detection has a great impact in providing the situational awareness increasing the maritime security.

Generally, automatic behavioral analysis methods are divided into three categories, statistical methods [4], [8], [11], [13], neural networks [2] and machine learning [10]. Mostly, these methods confronted with some drawbacks, such as high false alarm rates, over-fitting, ignoring some of the useful features of the vessels, the impossibility of generalizing some methods due to their specificity, unavailability of data and not covering all kinds of anomalies.

In this paper, we present a novel approach based on Hidden Markov Model (HMM) for detecting the anomalous behaviors of marine vessels. The proposed method consists of four steps including creating dataset, data processing, creating models and detecting anomalies. The experimental results of the proposed method on the dataset represent low False Alarm rate and its success in categorizing normal behaviors from abnormal behaviors. The rest of the paper is organized as follows. Section II presents the proposed method, section III focuses on the observations and experiences of implementing the proposed method. Finally, section IV draws conclusion.

II. PROPOSED METHOD

According to the above, in this section, a novel approach based on HMM has proposed to overcome the drawbacks of the previous works. The expected steps to reach the models of vessels' behaviors are as follows. The first step is to gather information and make a comprehensive database. Since the HMM requires features with discrete values, so it is necessary to discretize the features in the first step. Furthermore, the information provided in the first step usually contain raw data and needs to be refined, so in the second step, we must preprocess data for the implementation phase. In the third step, for each behavior of vessels, we construct an HMM and eventually, we will be able to classify the new unseen trajectories with the help of the constructed models. The behaviors of the vessels vary in each different sea routes. Therefore, different behavioral models must be constructed for each of the paths. DBSCAN algorithm can be used to

cluster marine traffic routes [1], [12]. In [4], a method has been proposed to detect the vessels activities on travelling to and departing from the harbor of Cape Town is provided by HMM, but the proposed method cannot be used to detect anomalies in other maritime routes. In [5], [6] a method based on HMM have been proposed to model animals behaviors that helped us how to use HMM and feature discretization in behavior modeling.

A. Creating Dataset

After selecting the corresponding route, values of the parameters speed, course, heading and coordination of the vessels have been collected in one week. Parameters, ΔS , ΔC , $Drift$ are derived from speed, course and the difference of course and heading respectively. ΔS is speed difference between two consecutive records of a vessel, and ΔC is calculated similarly. The discretization phase is done according to definitions of anomalies by an expert. After the discretization of the properties, label assignment for each discrete observation is shown in Table I.

B. Data Preprocessing

This phase consists of different steps. First, values of the parameters ΔC , $Drift$ and ΔS are calculated for each record. Therefore, observation vector will be, $O = [S, \Delta S, Drift, \Delta C]$. In the next step, we should classify the records into separate vessels to get trajectories. Given that, many of these points, due to their proximity, have no particular effect on the shape of the trajectory, if the number of points for each trajectory is high, some of them can be reduced. Another way to reduce the computational overload is removing stationary vessels.

C. Creating Models

An HMM with specified A,B and π parameters is shown as $\lambda = (A, B, \pi)$, where A is the transition probability distribution of states, B is the probability distribution of observations and π is probability distribution of start state. In HMM we assume that there is a set of states which are hidden, so the probability of the changing state will be hidden too, which causes the problem to be difficult. In the field of anomaly detection, hidden states may be due to lack of information or inability of understanding some facts. In this paper, three

hidden states are assumed, first, awareness of the physical and mental health of the person guiding vessel, second, environmental conditions such as other vessels' behaviors, atmosphere conditions and etc., third, vessels' operation and health.

There may be a state which is hidden from our sight, therefore, the problem will be modeled both with three and four hidden states. Finally, the model that classifies data more accurately will be selected. In this paper, a novel approach based on the HMM is presented to detect normal behaviors from four types of abnormal behaviors. Each behaviors relation with features is presented in Table II. In the proposed method, if there are k classes, then k HMMs must be constructed and trained. As a result, we have four HMMs for abnormal behaviors and one HMM for normal behavior, which uses the Baum-Welch algorithm for training each HMM and calculating its parameters [3]. Two stop criterions are considered for the Baum-Welch algorithm; First, the number of iterations is assumed 500 and the second, the convergence threshold is log likelihood that we assumed $1e-6$.

D. Detecting Anomalies

After constructing and training the models, to classify a new trajectory, we must calculate $P(\text{HMM}|\text{UnseenTrajectory})$ for each HMM using Forward Algorithm [3]. According to Fig. 1, P_n is observation probability produced by $P(\text{HMM}_n|\text{UnseenTrajectory})$ and P_6 is maximum of probabilities produced by abnormal behavior models HMM1 to HMM4. Finally, if P_6 is greater than P_5 the trajectory is classified as an anomaly, otherwise, it will be classified as normal behavior.

Model No.	Model name	Features related to the model
HMM1	Unexpected Stop	$S, \Delta S$
HMM2	Risky Speed	S
HMM3	Drift	$S, Drift$
HMM4	Spiral Movement	$S, \Delta C$
HMM5	Normal Behavior	$S, \Delta S, Drift, \Delta C$

TABLE II
RELATIONS OF EACH MODEL TO OBSERVATIONS

Feature	Observations	Label
S	Stopped	A
	Low Speed	B
	Medium Speed	C
	High Speed	D
ΔS	Low-speed Alterations	E
	Medium-speed Alterations	F
	High-speed Alterations	G
$Drift$	Low Drift	H
	High Drift	I
ΔC	Low-course Alterations	J
	High-course Alterations	K

TABLE I
OBSERVATION LABELING AFTER DISCRETIZATION

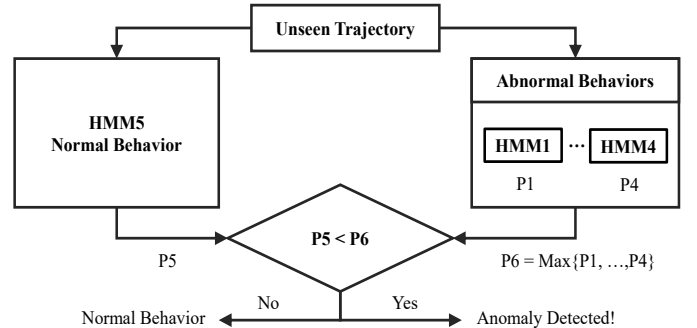


Fig. 1. Anomaly Detection Flowchart

III. RESULTS

A dataset consist of 201 vessels data sequence that has been labeled by an expert person is selected to test and train the model. In order to implement the proposed method, 70% of the data are selected randomly to train the models, and the other 30% include the test dataset. The HMM is defined with emission and transition matrices. For each of models, two matrices of transition and emission are calculated.

In this paper, *Precision*, *Recall*, *Accuracy* and *F – Measure* are calculated to evaluate and compare the proposed method with support vector machine classifier. Table III and IV show the experimental results of the classification of the data. According to the following tables, models constructed with three hidden states lead to higher accuracy in classification. Furthermore, the values of FN and FP are very low, indicating the objectives of the proposed method are satisfied. Also, the low value of FP indicates that a small number of vessels with normal behavior are classified incorrectly into an abnormal behavior class. Moreover, F_n=0 means that there is no mistake in classification of vessels with normal and abnormal behavior. This is very significant since no vessel with abnormal bahviour should be ignored.

Table V shows the performance results of the proposed method compared with the SVM classifier. The RBF core is used in the SVM algorithm for classification. The implementation results of the two methods show the high power of the proposed method in classification. There are two main reasons for explaining the high accuracy of the proposed method in comparison to the other. Firstly, the proposed method pays attention to the vessels navigation history, which is important in identifying marine behaviors. Most classifiers, including SVM, are not able to consider sequences of behavior in the classification. Secondly, marine behaviors are continuous and also similar to each other, so identifying the anomalies with the help of a single model is not adequate.

IV. CONCLUSION

A novel approach based on the HMM is introduced to identify the abnormal behaviors of marine vessels. Previous works included some defects such as high false alarm rates (FP), overfitting, non-coverage of different types of anomalies, the complexity of the methods, methods with special-purposes,

	<i>FN</i>	<i>FP</i>	<i>TN</i>	<i>TP</i>
3 Hidden States	0	2	14	45
4 Hidden States	0	3	13	45

TABLE III
TP, TN, FP, FN IN CLASSIFYING TEST DATA

	Models with 3 hidden states	Models with 4 hidden states
Accuracy	0.9672	0.9508
Precision	0.9574	0.9375
Recall	1	1
F-Measure	0.9782	0.9677

TABLE IV
EVALUATION CRITERIA VALUES IN CLASSIFYING TEST DATA

	Proposed method (HMM)	SVM
Accuracy	96.72%	85.25%
Precision	95.74%	70.85%
Recall	100%	75%
F-Measure	97.82%	72.72%

TABLE V
PROPOSED METHOD RESULTS COMPARED WITH SVM CLASSIFIER

etc. The results show that the proposed method performed much better than the SVM classifier. The strength of the proposed method from the analysis of the obtained results is high accuracy in classifying behaviors, low False Negative and False Positive rates.

ACKNOWLEDGMENT

We would like to express our special thanks of gratitude to JAKOTA Cruise System GmbH, referred to as FleetMon who supported us generously.

REFERENCES

- [1] L. Cazzanti and G. Pallotta, Mining maritime vessel traffic: Promises, challenges, techniques, *Proceedings of OCEANS*, (2015), 1–6.
- [2] V. Chandola, A. Banerjee, and V. Kumar. Anomaly detection: A survey. *ACM Computing Surveys (CSUR)*, **41**(3)(2009), 1-15.
- [3] R. Durbin, SR. Eddy, A. Krogh, G. Mitchison, Biological sequence analysis: probabilistic models of proteins and nucleic acids, Cambridge university press, (1998).
- [4] J. Du Toit, and J. H. Van Vuuren. Semi-automated maritime vessel activity detection using hidden Markov model, *Proceedings of the 43rd Annual Conference of the Operations Research Society of South Africa*, (2014), 71–78.
- [5] A. Franke, T. Caelli, and R. J. Hudson, Analysis of movements and behavior of caribou (*Rangifer tarandus*) using hidden Markov models, *Ecological Modelling*, **173**(2)(2004), 259–270.
- [6] Y. Guo, G. Poulton, P. Corke, G. J. Bishop-Hurley, T. Wark, D. L. Swain, Using accelerometer, high sample rate GPS and magnetometer data to develop a cattle movement and behaviour model, *Ecological Modelling*, **220**(17)(2009), 2068–2075.
- [7] S. Kazemi, S. Abghari, N. Lavesson, H. Johnson, and P. Ryman, Open data for anomaly detection in maritime surveillance, *Expert Systems with Applications*, **40**(14)(2013), 5719–5729.
- [8] R. Laxhammar, Anomaly detection for sea surveillance, 11th International Conference on Information Fusion, (2008), 1–8.
- [9] E. Martineau, J. Roy, Maritime anomaly detection: Domain introduction and review of selected literature, tech. rep., DEFENCE RESEARCH AND DEVELOPMENT CANADA VALCARTIER (QUEBEC), (2011).
- [10] C. Piciarelli, Micheloni, and G. L. Foresti, Anomalous trajectory patterns detection, 19th International Conference on Pattern Recognition, Tampa, FL, (2008), 1–4.
- [11] M. Smith, S. Reece, S. Roberts, et al, Maritime abnormality detection using Gaussian processes, *Knowledge and Information Systems*, **38**(3)(2014), 717–741.
- [12] F. Sun, Y. Deng, and F. Deng, Qingmeng Zhu and Hanyue Chu, Unsupervised maritime traffic pattern extraction from spatio-temporal data, 11th International Conference on Natural Computation (ICNC), (2015), 1218–1223.
- [13] Y. Zhou, W. Daamen, T. Vellinga, and S. P. Hoogendoorn, Vessel classification method based on vessel behavior in the port of Rotterdam, *Scientific Journals of the Maritime University of Szczecin*, **42**(114)(2015), 86–92.